

SKILL WEEK9

TASK1: List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State, Test Job Listings

```
laasya@laasya: ~/job_control_ x + v
laasya@laasya:~$ mkdir job_control_lab
laasya@laasya:~$ cd job_control_lab
laasya@laasya:~/job_control_lab$ sleep 60 &
[1] 1916
laasya@laasya:~/job_control_lab$ sleep 100 &
[2] 1919
laasya@laasya:~/job_control_lab$ sleep 120 &
[3] 1920
laasya@laasya:~/job_control_lab$ jobs
[1] Running sleep 60 &
[2]- Running sleep 100 &
[3]+ Running sleep 120 &
laasya@laasya:~/job_control_lab$ jobs -l
[1] 1916 Running sleep 60 &
[2]- 1919 Running sleep 100 &
[3]+ 1920 Running sleep 120 &
laasya@laasya:~/job_control_lab$ jobs -p
1916
1919
1920
laasya@laasya:~/job_control_lab$ jobs -r
[1] Running sleep 60 &
[2]- Running sleep 100 &
[3]+ Running sleep 120 &
laasya@laasya:~/job_control_lab$ kill %1
laasya@laasya:~/job_control_lab$ jobs
[1] Terminated sleep 60
[2]- Running sleep 100 &
[3]+ Running sleep 120 &
laasya@laasya:~/job_control_lab$ sleep 5 &
[4] 1926
laasya@laasya:~/job_control_lab$ jobs
[2] Done sleep 100
[3]- Done sleep 120
[4]+ Done sleep 5
laasya@laasya:~/job_control_lab$ |
```

TASK2: Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching

```
laasya@laasya: ~/job_control
laasya@laasya:~/job_control_lab$ jobs -r
[1]  Running          sleep 60 &
[2]-  Running          sleep 100 &
[3]+  Running          sleep 120 &
laasya@laasya:~/job_control_lab$ kill %1
laasya@laasya:~/job_control_lab$ jobs
[1]  Terminated       sleep 60
[2]-  Running          sleep 100 &
[3]+  Running          sleep 120 &
laasya@laasya:~/job_control_lab$ sleep 5 &
[4] 1926
laasya@laasya:~/job_control_lab$ jobs
[2]  Done              sleep 100
[3]-  Done              sleep 120
[4]+  Done              sleep 5
laasya@laasya:~/job_control_lab$ sleep 100 &
[1] 1934
laasya@laasya:~/job_control_lab$ jobs
[1]+  Running          sleep 100 &
laasya@laasya:~/job_control_lab$ fg %1
sleep 100
^Z
[1]+  Stopped           sleep 100
laasya@laasya:~/job_control_lab$ jobs
[1]+  Stopped           sleep 100
laasya@laasya:~/job_control_lab$ bg %1
[1]+ sleep 100 &
laasya@laasya:~/job_control_lab$ jobs
[1]+  Running          sleep 100 &
laasya@laasya:~/job_control_lab$ fg %1
sleep 100
laasya@laasya:~/job_control_lab$ jobs
laasya@laasya:~/job_control_lab$ sleep 200 &
[1] 1944
laasya@laasya:~/job_control_lab$ jobs -l
[1]+ 1944 Running       sleep 200 &
laasya@laasya:~/job_control_lab$ fg %1
sleep 200
^Z
[1]+  Stopped           sleep 200
laasya@laasya:~/job_control_lab$ |
```

TASK 3: Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_sigint(int sig)
{
    printf("\nSIGINT received!\n");
}

int main()
{
    sigset_t set;

    signal(SIGINT, handle_sigint);

    sigemptyset(&set);
    sigaddset(&set, SIGINT);

    printf("Blocking SIGINT for 10 seconds...\n");

    sigprocmask(SIG_BLOCK, &set, NULL);

    for (int i = 1; i <= 10; i++)
    {
        printf("Running... %d\n", i);
        sleep(1);
    }

    printf("\nUnblocking SIGINT now.\n");

    sigprocmask(SIG_UNBLOCK, &set, NULL);

    printf("Program continues.\n");

    sleep(5);

    return 0;
}
```

```

laasya@laasya:~/job_control_lab$ cd ~/job_control_lab
laasya@laasya:~/job_control_lab$ nano signal_mask.c
laasya@laasya:~/job_control_lab$ gcc signal_mask.c -o signal_mask
laasya@laasya:~/job_control_lab$ ./signal_mask
Blocking SIGINT for 10 seconds...
Running... 1
Running... 2
Running... 3
Running... 4
Running... 5
Running... 6
Running... 7
Running... 8
Running... 9
Running... 10
^C
Unblocking SIGINT now.

SIGINT received!
Program continues.
laasya@laasya:~/job_control_lab$ sleep 100
^Z
[2]+  Stopped                  sleep 100
laasya@laasya:~/job_control_lab$ jobs
[1]-  Stopped                  sleep 200
[2]+  Stopped                  sleep 100
laasya@laasya:~/job_control_lab$ bg %1
[1]-  sleep 200 &
laasya@laasya:~/job_control_lab$ ps -o pid,ppid,pgid,sid,tpgid,stat,cmd
  PID   PPID   PGID   SID   TPGID  STAT  CMD
  1893   1891   1893   1893   1964  Ss    -bash
  1963   1893   1963   1893   1964  T     sleep 100
  1964   1893   1964   1893   1964  R+    ps -o pid,ppid,pgid,sid,tpgid,s
[1]-  Done                    sleep 200
laasya@laasya:~/job_control_lab$ |

```

TASK 4: Resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery

```

laasya@laasya: ~/job_control
[2]+  Stopped                  sleep 100
laasya@laasya:~/job_control_lab$ jobs
[1]-  Stopped                  sleep 200
[2]+  Stopped                  sleep 100
laasya@laasya:~/job_control_lab$ bg %1
[1]-  sleep 200 &
laasya@laasya:~/job_control_lab$ ps -o pid,ppid,pgid,sid,tpgid,stat,cmd
  PID   PPID   PGID   SID   TPGID  STAT  CMD
  1893   1891   1893   1893   1964   Ss    -bash
  1963   1893   1963   1893   1964   T     sleep 100
  1964   1893   1964   1893   1964   R+    ps -o pid,ppid,pgid,sid,tpgid,s
[1]-  Done                    sleep 200
laasya@laasya:~/job_control_lab$ sleep 100
^Z
[3]+  Stopped                  sleep 100
laasya@laasya:~/job_control_lab$ jobs
[2]-  Stopped                  sleep 100
[3]+  Stopped                  sleep 100
laasya@laasya:~/job_control_lab$ bg %1
-bash: bg: %1: no such job
laasya@laasya:~/job_control_lab$ sleep 100
^Z
[4]+  Stopped                  sleep 100
laasya@laasya:~/job_control_lab$ jobs -l
[2]  1963 Stopped                  sleep 100
[3]- 1967 Stopped                  sleep 100
[4]+ 1976 Stopped                  sleep 100
laasya@laasya:~/job_control_lab$ kill -CONT 4567
-bash: kill: (4567) - No such process
laasya@laasya:~/job_control_lab$ fg %2
sleep 100
laasya@laasya:~/job_control_lab$ bg %2
-bash: bg: %2: no such job
laasya@laasya:~/job_control_lab$ jobs -l
[3]- 1967 Stopped                  sleep 100
[4]+ 1976 Stopped                  sleep 100
laasya@laasya:~/job_control_lab$ ps -ef | grep sleep
laasya      1967      1893   0 17:26 pts/2    00:00:00 sleep 100
laasya      1976      1893   0 17:29 pts/2    00:00:00 sleep 100
laasya      1980      1893   0 17:29 pts/2    00:00:00 grep --color=auto sleep
laasya@laasya:~/job_control_lab$ |

```